Clinical and Marketing Master Portfolio for Surgicover: Advanced Peri-Operative Clinical Nutrition

Section 1: B2B Company Brochure Content – Sparrow Pharmaceuticals' Peri-Operative Vision

The clinical management of surgical patients has undergone a structural paradigm shift over the last two decades. Traditional clinical practices, historically reliant on prolonged pre-operative fasting and reactive, delayed post-operative feeding, are increasingly being replaced by evidence-based, multimodal clinical pathways. Modern peri-operative frameworks, such as Enhanced Recovery After Surgery (ERAS), emphasize the critical importance of maintaining metabolic homeostasis, minimizing surgical stress, and initiating early nutritional rehabilitation. Surgical trauma inevitably induces a profound neuroendocrine and inflammatory stress response characterized by the hypersecretion of cortisol, catecholamines, and inflammatory cytokines. This systemic cascade drives a transient but severe state of peripheral insulin resistance and accelerates skeletal muscle proteolysis to mobilize amino acids for acute-phase protein synthesis and wound repair. Left unmanaged, this hypercatabolic state depletes glycogen reserves, degrades lean body mass, impairs diaphragmatic and myocardial function, and compromises the immune system. Consequently, malnourished or metabolically depleted patients face a three-fold increase in post-operative complications and a four-fold increase in mortality risk when undergoing major elective procedures.

To directly address these physiological vulnerabilities, Sparrow Pharmaceuticals has developed **Surgicover**, a highly specialized peri-operative clinical nutrition formulation designed to optimize patient recovery and facilitate accelerated hospital discharge. Surgicover represents a clinical solution built on the principles of metabolic prehabilitation and targeted immunonutrition. By delivering an easily digestible, macro- and micronutrient-optimized formula, Surgicover serves as an essential metabolic bridge throughout the surgical timeline.

The strategic inclusion of a rapid-acting protein matrix combined with specific free-form amino acids is clinically designed to shift the patient's metabolic balance from catabolic degradation to anabolic tissue synthesis. Furthermore, the complete elimination of added sucrose minimizes the risk of peri-operative glycemic fluctuations, a critical factor since acute post-operative hyperglycemia is directly correlated with increased surgical site infections, delayed wound closure, and prolonged hospital stays.

Implementing Surgicover within standardized clinical protocols allows healthcare institutions to improve clinical outcomes while enhancing operational efficiency. By supporting accelerated wound healing, preserving lean muscle tissue, and enabling early mobilization, Surgicover directly helps shorten the post-operative length of stay and reduce the incidence of preventable readmissions. This dual benefit optimizes both clinical recovery and institutional bed turnover, establishing Surgicover as an important component of modern peri-operative care.

Section 2: Product Leaflet – Technical Specifications

and Practical Guidance

Why Choose Surgicover?

Surgicover is a specialized, clinically designed nutritional supplement engineered to meet the elevated metabolic demands of patients undergoing major surgical interventions. The formulation delivers targeted support to accelerate tissue healing, preserve skeletal muscle mass, and optimize recovery pathways.

- **Dual-Action Amino Acid Spike:** Formulated with a targeted combination of free-form L-Arginine (1000\text{ mg}/100\text{ g}; 200\text{ mg}/\text{serve}) to accelerate wound healing and L-Leucine (500\text{ mg}/100\text{ g}; 100\text{ mg}/\text{serve}) to stimulate muscle protein synthesis.
- **Highly Bioavailable Hybrid Protein Matrix:** Utilizes a blend of Soya Protein Isolate, Defatted Soya Flour, and Skimmed Milk Powder, achieving a Protein Digestibility Corrected Amino Acid Score (\text{PDCAAS}) of 1.0 for optimal tissue remodeling.
- **Zero Added Sugar (Sucrose):** Completely free from added sucrose to support glycemic control during the post-operative stress phase, utilizing premium non-caloric sucralose to ensure patient compliance without compromising blood sugar stability.
- **Micro-Nutrient Synergy:** Packed with nine essential vitamins and five critical minerals, including high-dose Vitamin C (32.0\text{ mg}/\text{serve}) and Iron (2.40\text{ mg}/\text{serve}) to drive collagen cross-linking, oxygen transport, and energy production.
- **Dry Fruits Flavour:** A highly palatable, premium taste profile developed specifically to encourage compliance in patients experiencing post-operative nausea, taste alterations, or appetite suppression.

Instructions for Use

To prepare one serving of Surgicover:

1. Measure 150\text{ ml} of lukewarm water or milk into a clean container.
2. Add exactly one (1) heaped scoop (equivalent to 20\text{ g}) of Surgicover powder.
3. Stir or shake briskly until the powder has completely dissolved.
4. Consume immediately after preparation.

For optimal peri-operative metabolic support, it is recommended to consume one to two servings daily, or as directed by the attending clinical specialist.

Nutritional Composition Table

Nutrient Parameter	Unit of Measure	Declared Value Per 20\text{ g} Serve	Derived Value Per 100\text{ g} Powder
Energy	\text{Kcal}	75.64	378.20
Carbohydrates	\text{g}	15.94	79.70
Of which Added Sugars (Sucrose)	\text{g}	0.00	0.00
Protein (30% by weight)	\text{g}	6.00	30.00
Fat	\text{g}	0.19	0.95

Nutrient Parameter	Unit of Measure	Declared Value Per 20\text{ g} Serve	Derived Value Per 100\text{ g} Powder
L-Arginine	\text{mg}	200.00	1000.00
L-Leucine	\text{mg}	100.00	500.00
Vitamin B1 (Thiamine)	\text{mg}	0.60	3.00
Vitamin B2 (Riboflavin)	\text{mg}	0.73	3.65
Niacinamide (Vitamin B3)	\text{mg}	8.00	40.00
Vitamin B12 (Cyanocobalamin)	\text{mcg}	0.32	1.60
Pantothenic Acid (Vitamin B5)	\text{mg}	0.67	3.35
Vitamin D	\text{IU}	60.00	300.00
Vitamin B6 (Pyridoxine)	\text{mg}	0.32	1.60
Folic Acid (Vitamin B9)	\text{mcg}	152.00	760.00
Vitamin C (Ascorbic Acid)	\text{mg}	32.00	160.00
Manganese	\text{mg}	0.50	2.50
Iron	\text{mg}	2.40	12.00
Copper	\text{mg}	0.32	1.60
Calcium	\text{mg}	0.323	1.615
Phosphorus	\text{mg}	0.253	1.265

Ingredients: Soya Protein Isolate, Defatted Soya Flour, Skimmed Milk Powder, Added Amino Acids (L-Arginine, L-Leucine), Vitamins (Ascorbic Acid, Niacinamide, Riboflavin, Thiamine Mononitrate, Pyridoxine Hydrochloride, Folic Acid, Calcium D-Pantothenate, Cholecalciferol, Cyanocobalamin), Minerals (Ferrous Ascorbate, Manganese Sulfate, Copper Sulfate, Tribasic Calcium Phosphate), Sweetener (Sucralose), Permitted Flavours (Dry Fruits).
Regulatory Declarations: CONTAINS NATURALLY OCCURRING SUGARS. Contains Milk and Soy allergens. Not for parenteral/intravenous use. Best stored in a cool, dry place away from direct sunlight.

Section 3: Patient Diet Sheets – Integrating Surgicover into Recovery Pathways

Maximizing the therapeutic benefits of Surgicover requires systematic integration into the patient's daily diet throughout the peri-operative timeline. The following structured schedules outline the application of Surgicover from the pre-operative prehabilitation phase through the post-operative transition from liquids to solid foods.

Phase I: Pre-Operative Nutritional Prehabilitation (7 to 5 Days Pre-Surgery)

The goal during this phase is to build protein reserves and optimize micronutrient levels prior to the metabolic stress of surgery. Patients should consume a nutrient-dense, balanced diet rich in whole grains, lean proteins, and vegetables, supplemented with Surgicover to build systemic nitrogen balance.

Time of Day	Dietary Framework	Surgicover Integration Strategy
Breakfast	Whole-grain oatmeal or ragi porridge prepared with skimmed milk or water.	None.
Mid-Morning	Clear vegetable soup, a handful of almonds, or fresh whole fruit.	**Surgicover Serving 1:** 1 serving (20\text{ g} dissolved in 150\text{ ml} of lukewarm water).
Lunch	Lean chicken breast, baked fish, or stewed lentils paired with steamed vegetables and brown rice.	None.
Mid-Afternoon	Light buttermilk, green tea, or sprouted mung bean salad.	None.
Dinner	Steamed vegetables with paneer, tofu, or lean fish; whole-wheat flatbread or quinoa.	None.
Before Bed	Warm water or decaffeinated chamomile tea.	**Surgicover Serving 2:** 1 serving (20\text{ g} dissolved in 150\text{ ml} of skimmed milk).

Phase II: The Pre-Operative Window (24 Hours to 2 Hours Pre-Anesthesia)

During this phase, diet modifications help minimize the risk of pulmonary aspiration while avoiding the insulin resistance and catabolism caused by prolonged fasting.

- **24 Hours to 6 Hours Pre-Surgery:** Consuming a light, low-residue dinner (such as clear broths, plain rice, or soft semolina) is permitted. Solid foods must be discontinued 6 hours prior to the scheduled surgery.
- **6 Hours to 2 Hours Pre-Surgery:** Clear liquids should be consumed to support pre-operative hydration. Under explicit clinical supervision, a clear carbohydrate-rich beverage should be taken up to 2 hours before anesthesia to stimulate insulin secretion, preserve glycogen stores, and reduce post-operative insulin resistance.
- **Within 2 Hours of Surgery:** Nil-by-mouth (NPO) must be strictly enforced. Surgicover, due to its protein and fat content, must not be consumed during this immediate pre-anesthetic window to ensure safe gastric emptying.

Phase III: Post-Operative Transitional Feeding (Day 1 to Day 14 Post-Surgery)

Post-operatively, nutrition support aims to meet protein requirements (1.2\text{ g/kg/day} to 2.0\text{ g/kg/day}) to drive tissue repair, limit muscle loss, and support early mobilization.
[Post-Op Day 1: Clear Liquids] ──> [Post-Op Days 2-5: Soft/Pureed Foods] ──> [Post-Op Days

6-14+: Solid Foods]

Clear broths & water (No Surgicover yet)
Surgicover in water (2 servings per day)
Surgicover in milk/water
[span_23](start_span)[span_23](end_span) (2 servings per day)

- **Phase A (Typically Post-Operative Day 1):** Early oral feeding is initiated with clear liquids such as plain water, coconut water, or clear strained vegetable broth as tolerated.
- **Phase B (Typically Post-Operative Days 2 to 5):** As the patient transitions to a soft or pureed diet, Surgicover is introduced to prevent nitrogen deficit. Patients should consume soft, easily digestible foods like plain yogurt, mashed bananas, khichdi, or pureed lentils.
  - *Surgicover Protocol:* Take 1 serving of Surgicover ([span_90](start_span)[span_90](end_span)[span_93](start_span)[span_93](end_span)20\text{ g}) dissolved in 150\text{ ml} of lukewarm water twice daily—once in the mid-morning and once in the evening—to provide continuous amino acid availability.
- **Phase C (Typically Post-Operative Days 6 to 14 and Beyond):** The patient transitions back to a standard solid diet, with a strong focus on high-protein intake to support tissue remodeling and strength recovery.
  - *Surgicover Protocol:* Continue taking 1 to 2 servings of Surgicover daily. The powder can be dissolved in lukewarm milk or water, or blended into pureed oatmeal or warm porridges to enhance compliance during extended recovery.

Section 4: Department-Specific Clinical Paper/Whitepaper – The Science of Targeted Immunonutrition

Abstract

Peri-operative metabolic stress triggers catabolic cascades that degrade skeletal muscle tissue and impair wound healing. This whitepaper examines the biochemical mechanics of Surgicover, a peri-operative nutritional supplement formulated with a dual-action amino acid spike of L-Arginine and L-Leucine in a zero-added-sucrose hybrid protein matrix. Clinical evidence demonstrates that targeted L-Arginine supplementation upregulates nitric oxide synthesis and collagen deposition, significantly improving wound-breaking strength. Concurrently, L-Leucine acts as a metabolic signal to activate the mammalian target of rapamycin complex 1 (\text{mTORC1}) pathway, overcoming post-surgical anabolic resistance and preserving lean skeletal mass. Utilizing a high-bioavailability protein blend with a Protein Digestibility Corrected Amino Acid Score (\text{PDCAAS}) of 1.0, Surgicover provides a reliable clinical option to minimize surgical complications and accelerate recovery within Enhanced Recovery After Surgery (\text{ERAS}) protocols.

Introduction: The Metabolic Consequences of Surgical Stress

Surgical interventions, though therapeutic, impose a acute physiological stress load on the patient. This stress triggers a highly coordinated neuroendocrine response characterized by the activation of the hypothalamic-pituitary-adrenal axis and the sympathetic nervous system. The resulting surge in cortisol, glucagon, and catecholamines induces a hypercatabolic state, driving glycogenolysis, lipolysis, and severe skeletal muscle proteolysis.

In parallel, surgical trauma initiates an inflammatory response, releasing pro-inflammatory cytokines such as tumor necrosis factor-alpha ($\text{TNF-}\alpha$), interleukin-1 (IL-1), and interleukin-6 (IL-6). These cytokines alter peripheral insulin receptor signaling, creating transient insulin resistance that leads to post-operative hyperglycemia and impaired cellular glucose uptake.

Under these stress conditions, the body’s basal metabolic rate increases, while its ability to utilize standard nutritional substrates is compromised. If nutritional support is delayed, the patient experiences rapid depletion of functional protein reserves, directly undermining the structural integrity of visceral organs, skeletal muscle, and wound tissue. Addressing this hypercatabolic cascade requires targeted, highly bioavailable immunonutrition engineered to modulate immune responses, support protein synthesis, and maintain glycemic control.

Biochemical Mechanics of the L-Arginine Collagen Pathway

L-Arginine is classified as a conditionally essential amino acid; while healthy individuals synthesize sufficient quantities endogenously, surgical trauma, major hemorrhage, and septic shock rapidly deplete systemic reserves. During these physiological stress states, cellular demand for L-Arginine far outpaces the body's biosynthetic capacity, necessitating exogenous supply. L-Arginine serves as the sole precursor for nitric oxide (NO) synthesis, a reaction catalyzed by the nitric oxide synthase (NOS) enzyme family:

$$\text{L-Arginine} + \text{O}_2 + \text{NADPH} \xrightarrow{\text{NOS}} \text{L-Citrulline} + \text{NO} + \text{NADP}^+$$

At the wound site, nitric oxide acts as a key orchestrator of the tissue repair process. It regulates local microvascular blood flow, ensuring the delivery of oxygen, inflammatory cells, and essential nutrients to the hypoxic wound environment. Nitric oxide also modulates macrophage and T-lymphocyte activity, helping to control local wound infections.

Beyond the NOS pathway, L-Arginine is a vital substrate for the arginase enzyme, which converts L-Arginine into L-Ornithine. L-Ornithine is subsequently metabolized via the ornithine decarboxylase pathway to produce polyamines, which are essential for cell proliferation and tissue growth, and via the ornithine aminotransferase pathway to synthesize L-Proline. L-Proline is a critical structural precursor for collagen, the primary protein responsible for wound matrix stability and tensile strength.

Clinical and preclinical trials demonstrate that supplemental L-Arginine significantly enhances wound-breaking strength and collagen deposition. In trauma-hemorrhage and diabetic models, L-Arginine supplementation restored depressed levels of hydroxyproline (the index of new collagen synthesis) and upregulated mRNA levels for Type I and Type III procollagen. In clinical trials involving human volunteers, oral L-Arginine supplementation increased collagen deposition in standardized wounds by over 40%, alongside a significant increase in T-cell-mediated immune responses, directly supporting its role in reducing wound complications and dehiscence.

L-Leucine and the Overcoming of Post-Surgical Anabolic Resistance

The loss of skeletal muscle mass during the post-operative period is a primary driver of prolonged recovery, physical weakness, and functional decline. Post-operative immobilization, systemic inflammation, and inadequate protein intake combine to create a state of "anabolic resistance," where skeletal muscle tissue fails to respond to normal blood amino acid elevations. Under these conditions, standard protein intake is insufficient to stimulate muscle protein synthesis or halt proteolysis.

L-Leucine acts as a direct molecular trigger to overcome this anabolic resistance. Unlike most amino acids that undergo first-pass hepatic metabolism, branched-chain amino acids—particularly L-Leucine—bypass the liver and directly enter the systemic circulation, where they are taken up by skeletal muscle tissue. L-Leucine operates as an intracellular signaling molecule that activates the \text{mTORC1} pathway, a master regulator of cellular translation and protein synthesis.

Upon entering skeletal muscle cells, L-Leucine is sensed by intracellular mechanisms, triggering the activation of the GTPase-activating protein complex Sestrin2-GATOR2. This cascade recruits \text{mTORC1} to the lysosomal membrane, where it is activated. Active \text{mTORC1} phosphorylates two primary downstream targets:

1. **p70S6K (ribosomal protein S6 kinase beta-1):** Promotes translation initiation and ribosomal biogenesis.
2. **4E-BP1 (eukaryotic translation initiation factor 4E-binding protein 1):** Relieves inhibition on translation initiation factor complexes.

\text{L-Leucine} \longrightarrow \text{Sestrin2-GATOR2} \longrightarrow \text{mTORC1} \
\begin{cases} \xrightarrow{\oplus} \text{p70S6K} \longrightarrow \text{Protein Translation} \\
\xrightarrow{\ominus} \text{4E-BP1} \longrightarrow \text{Translation Initiation} \end{cases}

This activation directly initiates the translation of myofibrillar proteins, effectively stimulating muscle protein synthesis independently of overall systemic insulin or non-essential amino acid concentrations. Additionally, L-Leucine reduces muscle proteolysis by inhibiting ubiquitin-proteasome-mediated degradation, helping to maintain lean skeletal mass during periods of forced immobilization.

Clinical trials in elderly and sarcopenic cohorts demonstrate that a leucine-enriched nutritional supplement successfully stimulates postprandial muscle protein synthesis, preserving appendicular lean mass, improving handgrip strength, and increasing walking speed. Furthermore, leucine supplementation has been shown to improve respiratory muscle function, significantly increasing maximum static expiratory force, which is key to reducing post-operative pulmonary complications and helping patients regain independence.

Analysis of the Soya-Dairy Hybrid Protein Architecture

The core protein architecture of Surgicover is constructed as a precise hybrid of plant- and dairy-derived proteins: Soya Protein Isolate, Defatted Soya Flour, and Skimmed Milk Powder. This blend is engineered to optimize digestibility, extend systemic amino acid availability, and deliver a balanced amino acid spectrum.

Soya Protein Isolate undergoes extensive refining to remove fat and complex carbohydrates, resulting in a protein density exceeding 90\% by weight. It is highly digestible and scores a \text{PDCAAS} of 1.0, equivalent to animal-derived reference proteins like casein and egg white. Soya protein is also naturally rich in L-Arginine, providing a high concentration of this amino acid compared to pure whey or casein.

Skimmed Milk Powder provides a dairy-based protein matrix containing slow-release casein and rapid-release whey. This combination establishes an intermediate absorption profile. While pure whey protein creates a rapid, transient spike in blood amino acids, and pure casein digests

slowly over several hours, the hybrid soya-milk matrix in Surgicover delivers a sustained supply of essential amino acids to healing tissues.
Furthermore, standardized ileal digestibility (\text{SID}) studies demonstrate that while soy and milk proteins both exhibit high apparent ileal digestibility, combining them helps offset individual amino acid limitations. This synergistic approach ensures the formulation meets all physiological indispensable amino acid requirements for tissues undergoing active repair.

Impact on Clinical and Health Economic Outcomes

Integrating targeted nutritional supplements like Surgicover into standardized clinical pathways yields clear, measurable improvements in clinical recovery and health economic performance.

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[Targeted Nutrition Support]
          |
          ├─► Preserved Lean Muscle Mass ──► Improved Expiratory Force & Faster Mobilization
          ├─► Accelerated Wound Healing ───► Reduced Incisional Dehiscence & Infection Rates[span_253](start_span)[span_253](end_span)[span_254](start_span)[span_254](end_span)
          └─► Maintained Glycemic Stability ──► Decreas[span_214](start_span)[span_214](end_span)[span_217](start_span)[span_217](end_span)ed Post-Operative Hyperglycemic Complications
          |
          ▼
[Reduced Hospital Length of Stay & Lower Readmission Rates] (Enhanced Bed Turnover & Reduced Costs)
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Clinical trials evaluating nutrition-support interventions within pre- and post-operative protocols demonstrate a significant reduction in total daily oral calorie deficits and a corresponding decrease in post-operative length of stay. In major surgical cohorts, the application of high-protein, amino acid-supplemented oral nutrition supports several key outcomes:

- Decreases overall post-operative complication rates, including surgical site infections and anastomotic leaks.
- Reduces incidence of nosocomial infections due to enhanced T-lymphocyte and macrophage activity.
- Supports faster return of normal gastrointestinal motility and gut barrier function.
- Accelerates recovery of physical function and handgrip strength, facilitating earlier physical therapy compliance.

From a health economics perspective, reducing the post-operative length of stay by even one day yields substantial cost savings for healthcare systems. By mitigating expensive complications like wound dehiscence and pneumonia, Surgicover helps institutions maximize clinical resource efficiency, lower readmission rates, and optimize bed utilization.

Section 5: Clinician and Patient FAQs

Clinical and Scientific FAQs (Targeting Healthcare Professionals)

1. What is the clinical rationale for combining Soya Protein Isolate and Skimmed

Milk Powder instead of using pure Whey Isolate?

The soya-dairy hybrid matrix in Surgicover is designed to provide both rapid and sustained systemic amino acid delivery. While whey protein isolates are absorbed rapidly, they create a transient amino acid spike that may be cleared before target tissues can fully utilize them for sustained repair. Soya Protein Isolate behaves as an intermediate-rate protein, and the skimmed milk component provides slow-release casein. This ensures a prolonged plasma amino acid level to support continuous tissue synthesis. Furthermore, Soya Protein Isolate naturally contains high levels of L-Arginine, supplementing the free-form L-Arginine spike to optimize wound-healing pathways. #### 2. Why is Surgicover sweetened with Sucralose rather than natural sugars like Honey or Fructose? Surgical trauma induces a transient state of insulin resistance, making tight glycemic control critical to preventing post-operative complications. Formulations containing high-glycemic carbohydrates like sucrose, fructose, or honey can cause rapid blood glucose spikes, which are associated with increased risk of surgical site infections and delayed healing. Surgicover utilizes sucralose, an FSSAI-approved non-caloric sweetener, to provide a palatable taste profile without impacting the patient's glycemic response or glycemic index.

3. How does the concentration of L-Arginine in Surgicover compare to the levels used in clinical trials for wound healing?

Clinical trials evaluating the impact of oral L-Arginine on collagen synthesis and wound-breaking strength typically utilize doses ranging from 10\text{ g} to 30\text{ g} of free L-Arginine per day in healthy or severely traumatized cohorts. Surgicover delivers 200\text{ mg} of free-form L-Arginine per 20\text{ g} serving (1000\text{ mg}/100\text{ g}), which acts synergistically alongside the naturally occurring arginine in the Soya Protein Isolate base. This combined approach helps restore depleted plasma arginine levels to support tissue repair without risking the gastrointestinal discomfort or osmotic diarrhea sometimes associated with high-dose free-amino-acid boluses.

4. Can Surgicover be utilized in patients with mild to moderate chronic kidney disease (CKD)?

In patients with compromised renal function (CKD Stage 3 or higher), dietary protein intake must be carefully monitored to prevent accelerating uremic symptoms. While Surgicover contains high-quality protein (6.0\text{ g} per 20\text{ g} serving), the total daily protein intake must be factored into the patient's overall renal diet plan. Due to the presence of phosphorus (0.253\text{ mg}/\text{serve}) and calcium (0.323\text{ mg}/\text{serve}), usage in CKD patients should be guided by regular monitoring of serum electrolytes and renal function, under the direction of a clinical nephrologist or dietitian.

Regulatory and Compliance FAQs (Targeting Marketing and Brand Managers)

5. Why is Surgicover labeled as "Zero Added Sugar" rather than "Sugar-Free" under FSSAI regulations?

FSSAI regulations define strict standards for sugar claims. To carry a "Sugar-Free" label, a product must contain less than 0.5\text{ g} of total sugars per 100\text{ g} or per serving. Because Surgicover utilizes Skimmed Milk Powder as a core protein source, it inherently contains lactose, which is a naturally occurring milk sugar. Therefore, the product does contain sugars. Under FSSAI's "Non-Addition of Sugars" guidelines, the claim "Zero Added Sugar" or "No Added Sugar" is permitted because no monosaccharides or disaccharides (like sucrose or table sugar) have been added during manufacturing. However, the label must display the mandatory disclosure: "CONTAINS NATURALLY OCCURRING SUGARS".

6. Can the brand claim Surgicover is "Diabetic-Friendly" or "Low GI" on its packaging under FSSAI 2026 Audit standards?

No, the brand cannot place a "Diabetic-Friendly" or "Low GI" claim on the packaging unless it compiles a formal Technical Dossier containing a human-subject Glycemic Index Testing Report from an accredited, recognized laboratory. Under the FSSAI 2026 guidelines, theoretical calculations of glycemic index are no longer accepted during regulatory audits. Making unsubstantiated claims can result in major non-conformance citations, mandatory product recalls, or financial penalties. The marketing team should focus language on "Zero Added Sucrose" and "Sweetened with Sucralose" until formal human GI testing is completed.

7. Does the presence of soy protein require allergen declarations under Indian labeling laws?

Yes, under FSSAI's Packaging and Labelling Regulations, soy and milk are classified as major food allergens and must be declared prominently on the label. The product packaging must include a clear, legible allergen statement, such as: "Contains Milk and Soya". This statement is typically positioned immediately below the ingredient list in a distinct font to ensure patient safety and maintain full regulatory compliance.

Section 6: Product Comparison Sheets

To support clinical marketing efforts, the following comparison outlines the key differences between Surgicover and major clinical nutrition products available in the market.

Macro-Nutrient and Compositional Comparison

Compositional Parameter	Surgicover (Per 20\text{ g} Serving)	Ensure High Protein (Per 20\text{ g} Equivalent)	PentaSure HP (Per 20\text{ g} Equivalent)	Resource High Protein (Per 20\text{ g} Equivalent)
Energy (Kcal)	75.64	\sim 76.00	\sim 82.00	74.40
Protein Content (g)	6.00	6.00	8.60	8.[span_299](start_span)[span_299](end_span)40 - 9.00
Protein Sources	Soya Protein Isolate, Defatted	Milk Protein Concentrate, Soy	Whey Protein Concentrate.	Whey Protein Concentrate,

Compositional Parameter	Surgicover (Per 20\text{ g} Serving)	Ensure High Protein (Per 20\text{ g} Equivalent)	PentaSure HP (Per 20\text{ g} Equivalent)	Resource High Protein (Per 20\text{ g} Equivalent)
	Soya Flour, Skimmed Milk Powder.	Protein Isolate.		Skimmed Milk Powder.
Carbohydrates (g)	15.94	\sim 11.20	\sim 9.60	8.34
Added Sugars / Strategy	Zero Added Sucrose; Sweetened with Sucralose.	Lower Sucrose.	Contains Fructose.	Contains Maltodextrin.
Fat Content (g)	0.19	Low Fat.	Contains Corn Fat.	0.56
L-Arginine (mg)	20[span_285](start_span)[span_285](end_span)[span_287](start_span)[span_287](end_span)0[span_316](start_span)[span_316](end_span).00 (Free-form added)	None added.	Inherent only.	None added.
L-Leucine (mg)	100.00 (Free-form added)	None added.	Inherent only.	None added.
Vitamin C (mg)	32.00	\sim 10.00	\sim 16.00	4.80
Vitamin D (IU)	60.00	\sim 30.00	\sim 40.00	1.20
Calcium (mg)	0.323	\sim 120.00	\sim 100.00	100.00
Phosphorus (mg)	0.253	\sim 90.00	\sim 80.00	78.00
Iron (mg)	2.40	\sim 1.50	\sim 1.80	\sim 1.20

Key Product Differentiators and Clinical Value

Surgicover

- **Target Clinical Benefit:** Specifically formulated for peri-operative recovery, accelerated wound healing, and lean myofibrillar preservation.
- **Biochemical Advantage:** Direct stimulation of the collagen pathway via an added L-Arginine spike, and activation of muscle protein synthesis via free L-Leucine.
- **Glycemic Control:** Zero added sucrose prevents post-operative blood glucose spikes, making it an appropriate choice during acute surgical stress.

Ensure High Protein

- **Target Clinical Benefit:** Formulated for active adults age 40+ to support general bone and muscle strength.
- **Biochemical Advantage:** Provides standard, high-quality proteins without targeted free-form amino acid spikes.

- **Glycemic Control:** Contains some sucrose and maltodextrin; designed for general nutrition rather than acute glycemic control.

PentaSure HP

- **Target Clinical Benefit:** High-protein, high-calorie formula designed to support muscle building and general weight gain.
- **Biochemical Advantage:** Relies on whey protein concentrate to support recovery, but lacks targeted free-form amino acid additions.
- **Glycemic Control:** Utilizes fructose and maltodextrin as primary carbohydrate sources, which can impact glycemic index.

Resource High Protein

- **Target Clinical Benefit:** General adult protein deficiency, geriatric weakness, and immune support.
- **Biochemical Advantage:** Rich in whey protein to support digestion, but does not provide targeted therapeutic amino acid additions.
- **Glycemic Control:** Contains maltodextrin, which has a high glycemic index and can cause rapid blood glucose fluctuations in peri-operative patients.

Works cited

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